1.

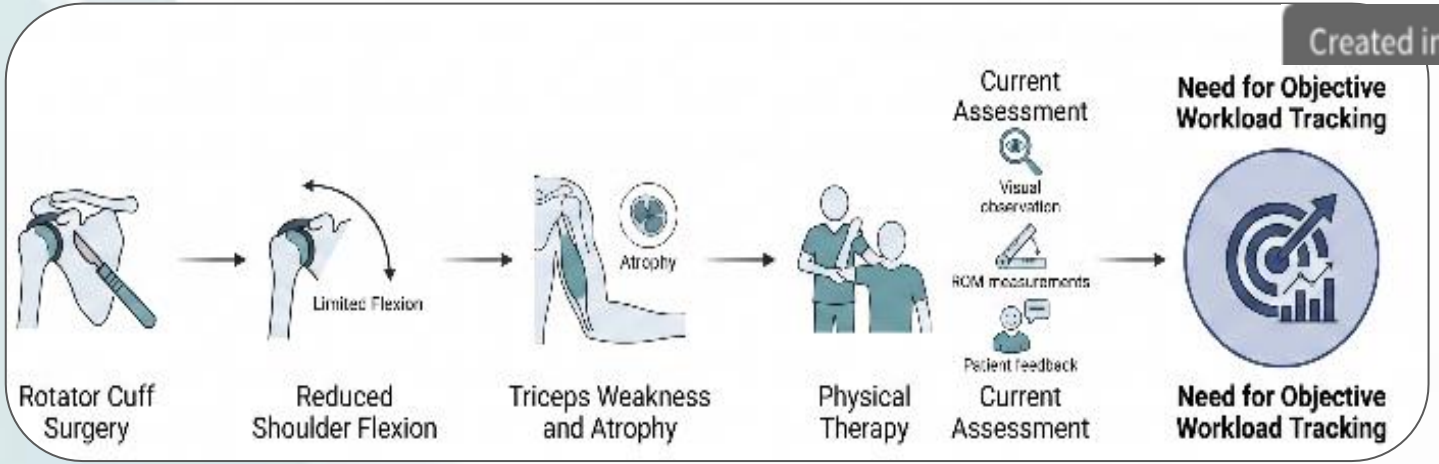
Quantifying Upper-Extremity Biomechanics Across Varying Shoulder Flexion Angles During Elbow Extension Using Low-Cost IMU and Computer Vision Tracking

Ethan Avila | Dr. Pranav Bhounsule | UIC SHARP!



2. Clinical Motivation

- **Rotator cuff surgery and elbow injuries** can lead to reduced shoulder range of motion and muscle weakness.
- Rehabilitation should gradually increase **shoulder flexion angle** to restore strength and range of motion (Littlewood et al., 2021).
- **How does shoulder flexion angle influence the estimated external mechanical workload during elbow-extension rehabilitation exercises?**



3.

Current Study Approaches



Motion Capture

- **High Accuracy**
- Expensive
- Bound to Laboratories
- Limited Accessibility



Clinical Assessment & Traditional Rehabilitation

- **Low Cost**
- **Widely Available**
- Limited Biomechanical Data
- Limited Quantitative Analysis

My Measurement Approach

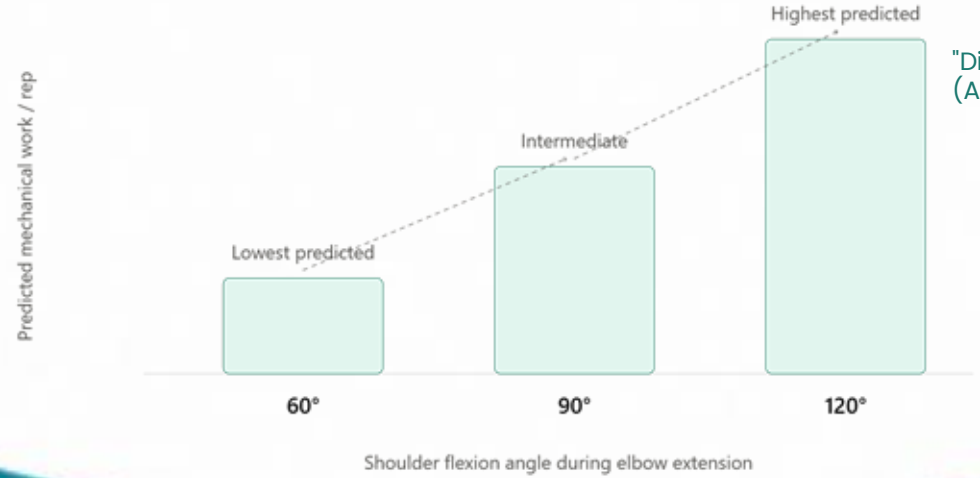


Computer Vision System & Wearable IMUS

- **Portable & Low Cost**
- **Prescribed Angle Tracking**
- **Mechanical Workload Estimation**
- **Quantitative Movement Data Analysis**

4. Hypothesis

- **If participants perform elbow-extension exercises at increasing shoulder flexion angles of 60°, 90°, and 120°, there will be an increased workload when measured by IMU's and Computer Vision.**



"Diagram generated with Claude (Anthropic)"

5. Study Design:

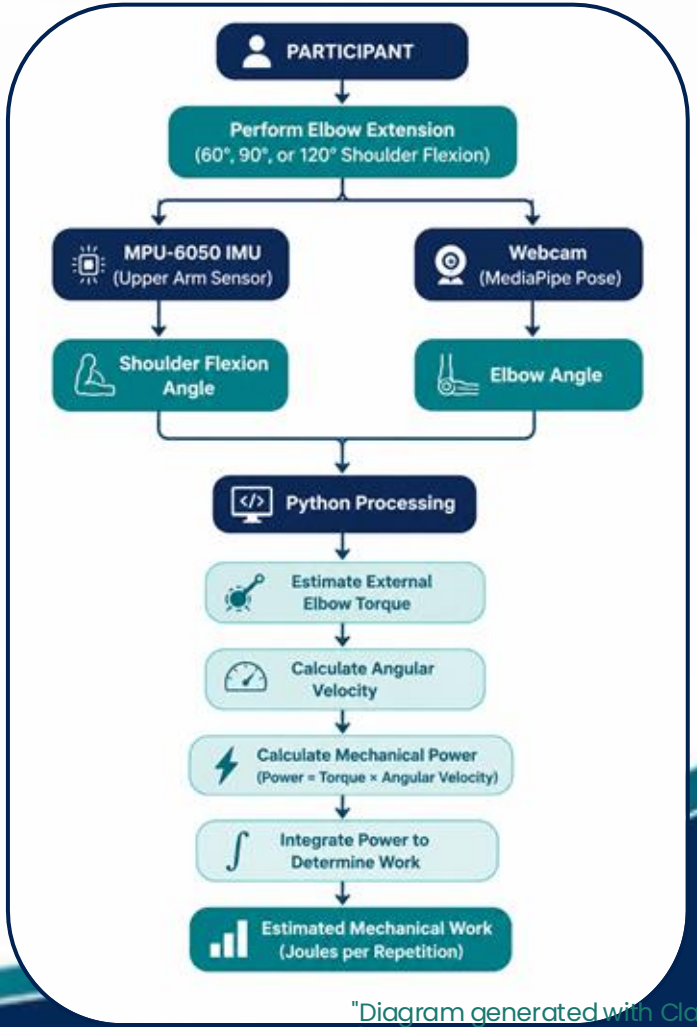
Category	Description
Participants	Healthy volunteers
Exercise	Dumbbell elbow extension
Shoulder angles	60°, 90°, 120°
Repetitions	5 repetitions per angle
Independent variable	Shoulder flexion angle
Dependent variable	Mechanical work per repetition
Constant factors during the experiment	Same weight, same exercise, same protocol

"Diagram generated with Claude (Anthropic)"

6. Methods:

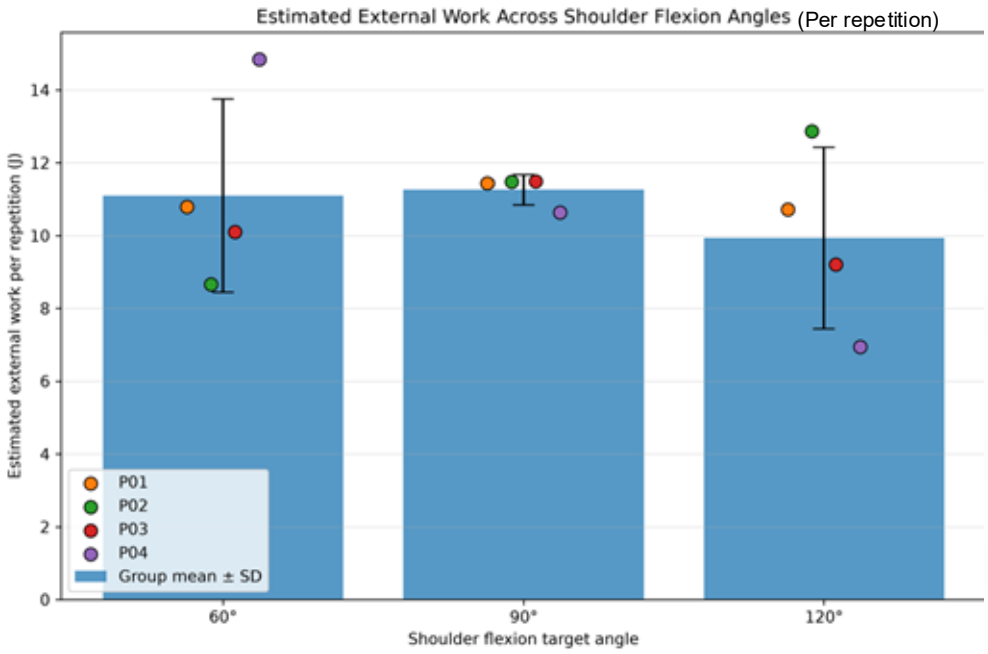






"Diagram generated with Claude (Anthropic)"

7. Results:



- Highest average workload observed at 90°
- Observed order: 90° > 60° > 120°
- The estimated mechanical workload varied across the tested shoulder flexion angles, although the observed trend differed from the original hypothesis."

- n = 4 participants
- Error bars = ±1 SD

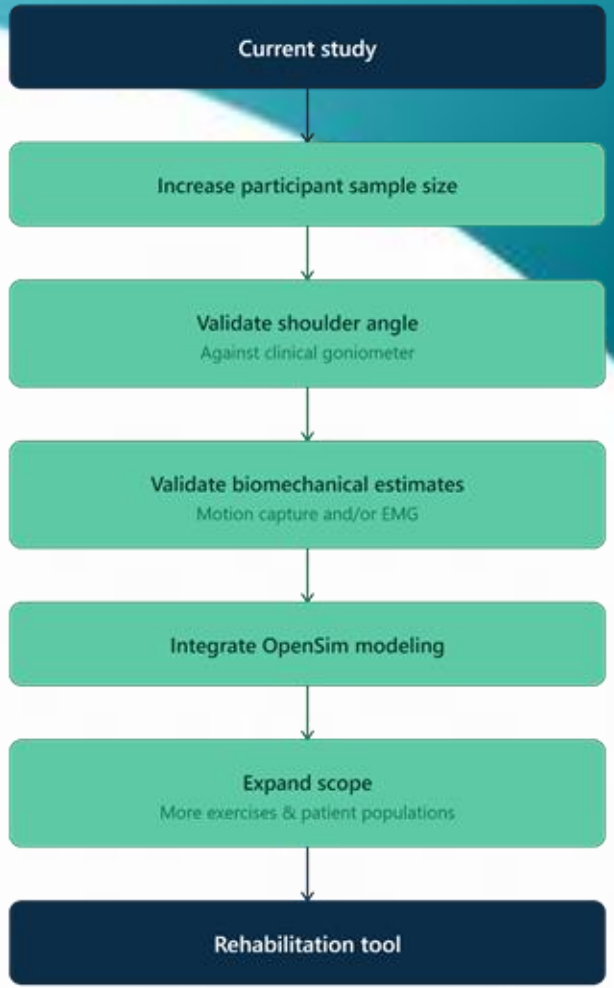
Conclusion

- This study investigated how **shoulder flexion angle influences estimated external mechanical workload** during elbow-extension rehabilitation exercises.
- Although the results did not support the original hypothesis..."
  - The low-cost IMU and computer vision system **successfully quantified upper-extremity movement** and estimated external mechanical workload.

9.

Future Direction and Application

- Increase **sample size**
- Validate shoulder angle using a **clinical goniometer**
- Validate biomechanical estimates with **motion capture and/or an EMG**
- Integrate OpenSim
- Expand to additional **rehabilitation exercises.**



"Diagram generated with Claude (Anthropic)"

10. Acknowledgements, Thank you to:

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All SHARP! Students

